Australian Journal of Earth Sciences (2011) **58**, (347–360)

Paleomagnetism of the Pandurra Formation and Blue Range Beds, Gawler Craton, South Australia, and the Australian Mesoproterozoic apparent polar wander path

P. W. SCHMIDT[1]* AND G. E. WILLIAMS[2]

[1]*CSIRO Earth Science & Resource Engineering, PO Box 136, North Ryde, NSW 1670, Australia.*
[2]*Discipline of Geology and Geophysics, School of Earth and Environmental Sciences, University of Adelaide, SA 5005, Australia.*

The Mesoproterozoic Pandurra Formation and Blue Range Beds of the Gawler Craton, South Australia, are undeformed, low-dipping (typically 10–15°) arenites of fluvial origin displaying pale to moderate red colours with light grey mottles and streaks and local Liesegang banding. Eighty-nine core samples and seven block samples were collected from 18 sites in the Pandurra Formation at Whyalla and 70 core samples from 16 sites in the Blue Range Beds at Talia. Specimens from all samples were stepwise thermally demagnetised, revealing hematite as the dominant magnetic carrier with a Neél temperature of 680°C. The mean direction for the Pandurra Formation after tilt-correction ($N = 90$ specimens) is declination = 248.2°, inclination = 46.7° ($\alpha_{95} = 4.4°$), indicating a paleolatitude of 27.9 +3.9/–3.4° and a pole position at latitude 33.6°S, longitude 64.5°E ($A_{95} = 4.9°$). The mean direction for the Blue Range Beds after tilt-correction ($N = 61$) is declination = 233.9°, inclination = 50.2° ($\alpha_{95} = 4.0°$), with a paleolatitude of 31.0 +3.7/–3.5° and a pole position at 45.2°S, 59.0°E ($A_{95} = 4.7°$). Combined results for the two units after tilt-correction ($N = 151$) is declination = 242.6°, inclination = 48.3° ($\alpha_{95} = 3.1°$), indicating a mean paleolatitude of 29.3 ± 2.8° and a mean pole position at 38.4°S, 62.4°E ($A_{95} = 3.5°$). The combined results constitute a positive fold-test with 99% confidence, suggesting that the magnetisation pre-dates tilting. The two pole positions, when compared with Australia's Mesoproterozoic apparent polar wander path, imply ages between 1.5 and 1.2 Ga, in accord with the inferred stratigraphic ages. The pole paths for Australia and North America are sub-parallel from about 1.4 to 1.2 Ga, before diverging. However, during this interval Australia and North America were widely separated, seemingly ruling out their assemblage in some supercontinental reconstruction for that interval.

KEY WORDS: South Australia, Gawler Craton, Mesoproterozoic, red beds, paleomagnetism, polar wander.

INTRODUCTION

The late Mesoproterozoic to early Neoproterozoic apparent polar wander path (APWP) for Australia is poorly defined, with few Australian rock units between 1.5 and 0.8 Ga yielding reliable paleomagnetic results (Idnurm & Giddings 1988; Idnurm 2000). The Mesoproterozoic Pandurra Formation and Blue Range Beds are important fluvial red bed units of the pre-Adelaidean succession in South Australia (Preiss 1987a, 1993), and although not well dated offer the opportunity to begin to fill this gap. Furthermore, testing their correlation through paleomagnetism may be relevant to the exploration for possible associated mineral deposits, for example unconformity-related uranium (Flint & Rankin 1991; Wilson *et al.* 2010). Here, we provide the first paleomagnetic study of these two units and discuss their place in the Proterozoic APWP for Australia.

STRATIGRAPHY AND GEOCHRONOLOGY

Pandurra Formation

The Pandurra Formation of the Cariewerloo Basin in the northeastern Gawler Craton (Figures 1, 2) comprises unmetamorphosed, low-dipping (typically ~15°) fluvial deposits (Cowley 1993; Parker & Fanning 1998). The formation is conglomeratic at the base and passes upward to mostly coarse-grained, poorly sorted, lithic and feldspathic sandstones with conglomerate bands and local shale and siltstone. Cross-bedding is common. The sandstones are pale to moderate red with numerous very light grey streaks and mottles (Figure 3a) (colour designations from the Geological Society of America Rock-Color Chart, 1963). The formation in the Whyalla 1:250 000 sheet area attains a thickness of 200+ m and in the northwestern Cariewerloo Basin is up to 1600 m

*Corresponding author: phil.schmidt@csiro.au

ISSN 0812-0099 print/ISSN 1440-0952 online © 2011 Geological Society of Australia
DOI: 10.1080/08120099.2011.570377

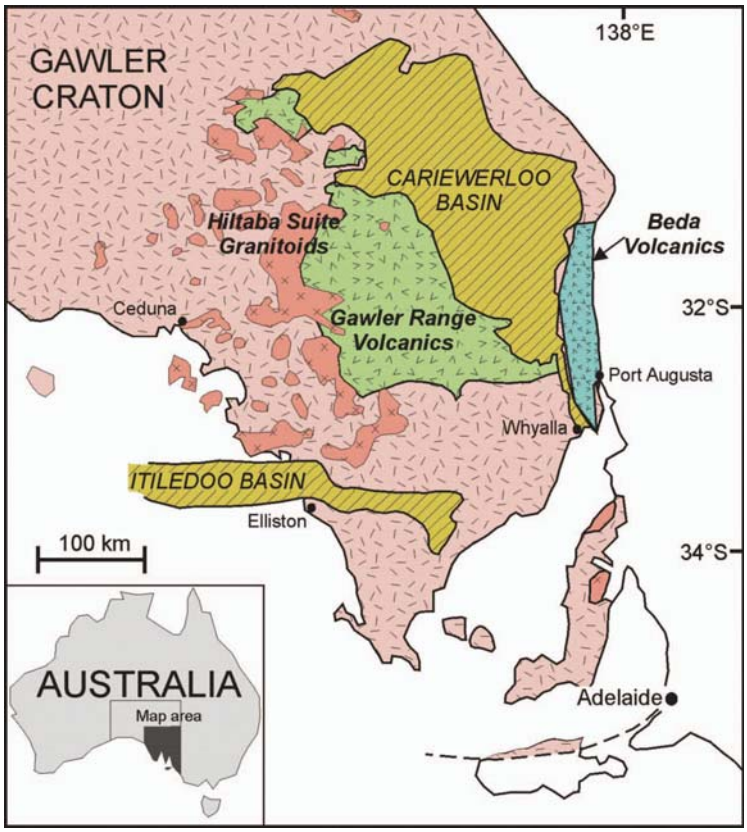


Figure 1 Mesoproterozoic tectonic features of central South Australia (adapted from Flint *et al.* 1993, figure 5.1).

thick. Silicification is common near the upper unconformable boundary, suggesting paleoweathering (Preiss 1987b). Liesegang banding of reddish pigmentation also occurs locally, suggesting fluid movement possibly during diagenesis; the presence of quartzite clasts with Liesegang banding in Sturtian age (Cryogenian) diamictites (Preiss 1987b) implies these processes were pre-Sturtian (>700 Ma).

The age of the Pandurra Formation is only loosely constrained. The formation overlies the Gawler Range Volcanics and Hiltaba Suite granitoids and hence is younger than 1.59 Ga (Fanning *et al.* 1988) (Table 1). The age of the overlying Beda Volcanics also is poorly constrained, with Rb–Sr isochron ages for basalt and dolerite of around 1.2–1.1 Ga (Cowley & Flint 1993) and a Sm–Nd isochron age of *ca* 800 Ma for the Gairdner dyke swarm which intrudes the Pandurra Formation (Zhao *et al.* 1994). Fanning *et al.* (1983) interpreted a whole-rock siltstone Rb–Sr isochron age of 1424 ± 51 Ma for the Pandurra Formation as a maximum age for deposition. Fanning & Link (2003, 2004) determined SHRIMP U–Pb ages of detrital zircons from the Pandurra Formation: a 1620–1580 Ma component formed 40–60% of the population, with older groups from 1780–1640 Ma and a subordinate group >2440 Ma. They concluded that the nearby 1595–1585 Ma Gawler Range Volcanics and Hiltaba Suite dominate the detrital zircon population. Available data thus are consistent with a Mesoproterozoic age for the Pandurra Formation.

Blue Range Beds

The fluvial Blue Range Beds of the Itiledoo Basin crop out sporadically across the central Eyre Peninsula (Figures 1, 2). The unit is unmetamorphosed and generally low-dipping (typically $\sim 15^{\circ}$), and comprises a basal pebble conglomerate, followed by coarse-grained, poorly sorted, lithic and feldspathic sandstones (Figure 3b) and conglomerate bands (Flint & Parker 1981; Flint & Rankin 1991; Flint 1992, 1993). Trough and planar cross-bedding is very common. Drilling indicates the Blue Range Beds are at least 401 m thick, and seismic profiles suggest a total thickness near 2500 m (Flint 1992). The sandstones show pale to moderate red colours with very light grey mottles and streaks, and some Liesegang banding. Local dewatering conduits also are very light grey (Figure 3c), suggesting that the reduction and removal of some hematitic pigment took place during compaction and early diagenesis; comparable early diagenetic mobilisation of Fe and mottling has been noted in other Proterozoic red beds (Williams 2001).

The age of the Blue Range Beds, like that of the Pandurra Formation, is only loosely constrained. The Blue Range Beds unconformably overlie the early Proterozoic Hutchison Group, therefore post-dating the *ca* 1.85–1.6 Ga Kimban Orogeny (Webb *et al.* 1986; Parker 1993; Zang & Fanning 2001). The conglomerates contain rhyolite clasts comparable with lithologies for the 1.59

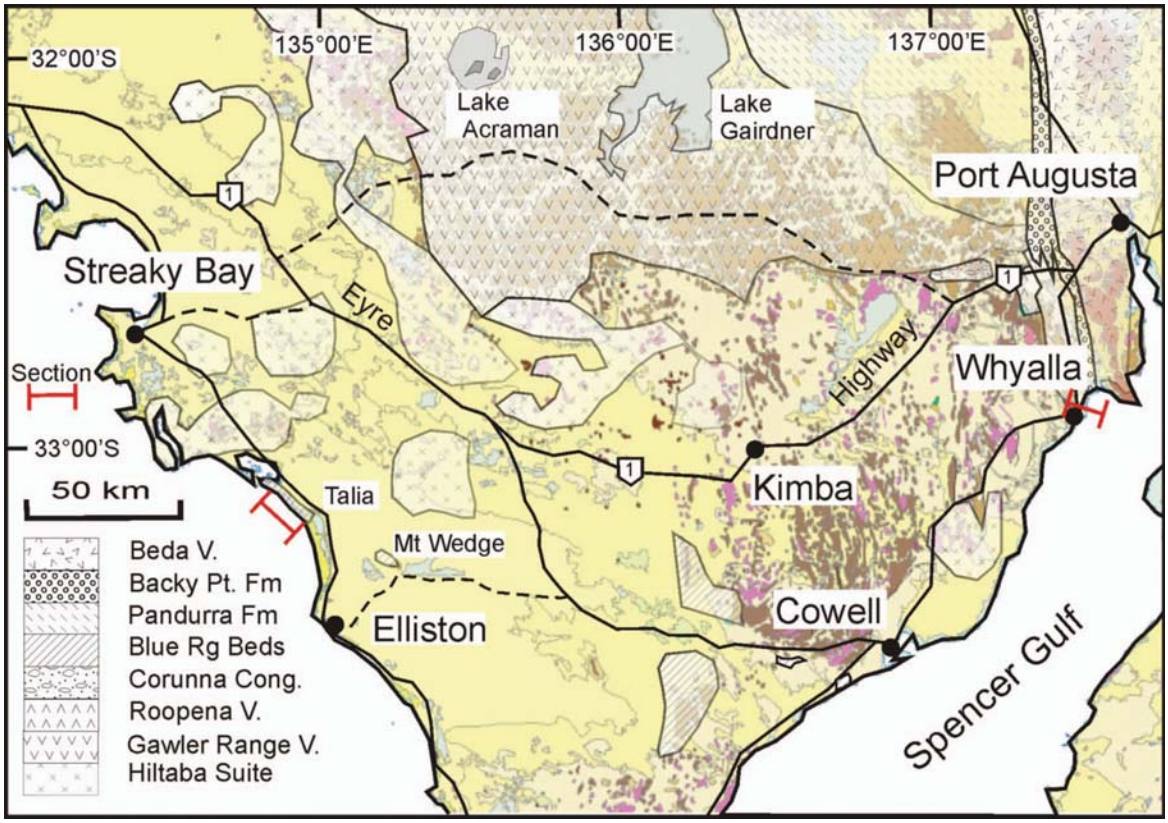


Figure 2 Geological map highlighting the Mesoproterozoic geology of the southern Gawler Craton–Eyre Peninsula and showing the location of sampled sections in the Pandurra Formation and Blue Range Beds; site details are given in Appendix 1. Adapted from Surface Geology of Australia 1:1 000 000 scale, South Australia © Commonwealth of Australia (Geoscience Australia) 2010, URL: <http://www.ga.gov.au/ausgeonews/ausgeonews200809/productnews.jsp#product1> (released under the Creative Commons Attribution 2.5 Australia Licence).

Ga Gawler Range Volcanics to the north, which the Blue Range Beds therefore also appear to post-date. Reported local folding with dips of up to 70° (Flint & Parker 1981; Flint & Rankin 1991) suggested deposition prior to the *ca* 1.5–1.4 Ga Wartakan Event (Flint & Parker 1981; Parker & Lemon 1982; Webb *et al.* 1986). However, we examined the Rudall area where relatively steep dips are shown (Flint & Rankin 1989) hoping to sample for a fold test on the Blue Range Beds, but found dips $\leq$35° and no evidence of tight folding. We therefore conclude that deposition of the Blue Range Beds cannot be inferred to pre-date the Wartakan Event. Flint & Parker (1981) suggested that the Blue Range Beds are correlative with the Corunna Conglomerate, to the north (Figure 2), which likewise comprises unmetamorphosed conglomerates and sandstones. Flint & Rankin (1991) suggested that correlation with the Pandurra Formation is possible, and Flint (1992) suggested that deposition of the Blue Range Beds was synchronous with, or followed intrusion of, the Hiltaba Suite.

Available stratigraphic evidence and geochronological data therefore imply that the Blue Range Beds and Pandurra Formation, although not necessarily correlative, are younger than *ca* 1.6 Ga and probably older than *ca* 1.1 Ga, i.e. Mesoproterozoic. The time of tilting of the Pandurra Formation and Blue Range Beds is unclear, but some tilting may have accompanied faulting, extension and basaltic volcanism in south-central Australia during early Adelaidean times when the *ca* 800 Ma Gairdner dyke swarm was intruded (Flint 1992; Zhao *et al.* 1994; Parker & Fanning 1998; Wingate *et al.* 1998).

PREVIOUS PALEOMAGNETIC STUDIES FOR THE AUSTRALIAN MESOPROTEROZOIC

Chamalaun & Dempsey (1978) reported paleomagnetic results for the Gawler Range Volcanics. Alternating field (AF) demagnetisation isolated magnetisation moderately upward-directed to the northeast. Thermal demagnetisation was performed on a few selected specimens representing different lithologies which showed a single magnetic carrier with a Curie (Neél) temperature below 600°C, consistent with that carrier being magnetite. The thermal results gave directions similar to those from the AF demagnetisation but no mean directions were reported. Chamalaun & Dempsey (1978) carried out XRD studies on representative lithologies and reported that only diffraction lines from magnetite were observed. The XRD data are thus consistent with the thermal demagnetisation results.

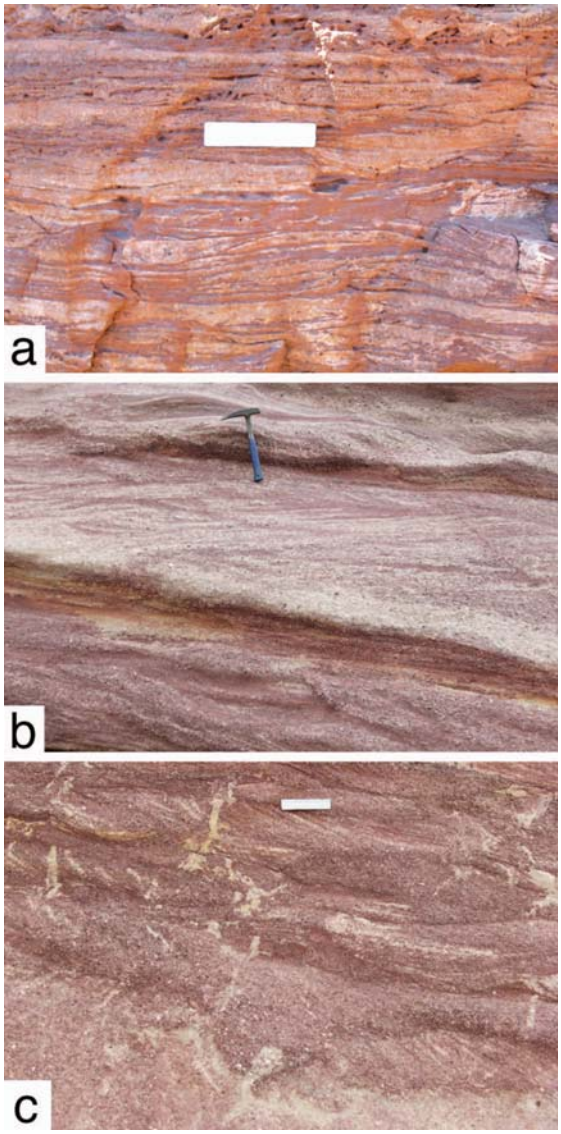


Figure 3 (a) Vertical section of cross-bedded coarse-grained sandstone of the Pandurra Formation, Freyers railway cutting, Whyalla (33°00′59″S, 137°33′13″E). Scale 15 cm. (b) Vertical section of cross-bedded coarse-grained sandstone of the Blue Range Beds, coastal exposure at The Monument near Talia, (33°19′39″S, 134°47′55″E). Hammer 33 cm long. (c) Vertical section of coarse-grained sandstone of the Blue Range Beds at The Monument near Talia (33°19′39″S, 134°47′55″E), showing very light grey, sub-vertical dewatering conduits. Scale 15 cm.

The mean direction determined was declination $D = 34.3°$, inclination $I = -48.1°$, $\alpha_{95} = 5.2°$ and corresponding pole position at latitude $\lambda_p = 60.4$ S, longitude $\phi_p = 50.0°$ E, $dp = 4.4°$, $dm = 6.8°$ (Chamalaun & Dempsey 1978). Directions of opposite polarity were not found. No field tests were performed to constrain the age of the magnetisation and although the pole position has been considered a 'key pole' for Australia at *ca* 1500 Ma (Idnurm & Giddings 1988), in light of a negative fold test and unstable nature of the remanence (see below) this status has been abandoned.

Schmidt & Clark (2011) have extended sampling of the Gawler Range Volcanics to include a steeply dipping section, near Uno, to allow the execution of a fold test. Although the age of tilting is unknown, the magnetisations were shown to post-date the tilting and were most likely overprints. Detailed rock magnetic measurement showed that the magnetic carriers were multi-domain magnetite grains, which were unlikely to retain original thermoremanent magnetisation. The probability that the Gawler Range Volcanics has been magnetically overprinted is enhanced by apatite fission track analysis that shows cooling of the Gawler Craton below the apatite fission track retention temperature (100 ± 20°C) at *ca* 300 Ma (Kohn *et al.* 2002). Apatite fission track analysis for shattered Yardea Dacite from the Acraman impact site yielded an age of 319 ± 19 Ma (Williams 1994). These Carboniferous ages are slightly younger than the Late Devonian age of the magnetic overprinting (Schmidt & Clark 2011), reflecting the somewhat higher blocking temperatures for the magnetic remanence.

Much paleomagnetic work has been undertaken on Paleoproterozoic and Mesoproterozoic successions in the McArthur Basin and Pine Creek Inlier and on the Lawn Hill Platform in northern Australia (Irving & Green 1958; Idnurm & Giddings 1988; Idnurm *et al.* 1995; Idnurm 2000). Although Irving & Green (1958) assumed the Nullagine Lavas (Pilbara) and Edith River Volcanics (Pine Creek Inlier) were Neoproterozoic, their paleomagnetic results are similar to more recent findings (see Schmidt & Embleton 1985; Strik *et al.* 2003) for the Fortescue Volcanics (which include the older termed Nullagine Volcanics) and by Idnurm & Giddings (1988) for the Plum Tree Creek Volcanics (which include the previously termed Edith River Volcanics). Idnurm & Giddings (1988) proposed a detailed pole path based on results from the *ca* 1.88 Ga Plum Tree Creek Volcanics to the 1.69–1.65 Ga Kombolgie Sandstone (McArthur Basin), and sequences in between. In addition, four phases of magnetic overprinting, OP1–OP4, were reported by Idnurm (2000). The OP1 and OP3 pole positions lie well away from the present magnetic pole, although the OP2 and OP4 poles are quite near it. Idnurm (2000) presented arguments involving statistical tests, including positive fold tests and negative tests for common mean directions, that he thought favoured a Precambrian origin for the associated magnetisations, but the possibility remains that they are in fact Tertiary and the tests are false positives (Schmidt & Williams 2008).

The status of the OP3 group of poles reflecting a period of overprinting in the McArthur Basin and on the Lawn Hill Platform is not disputed, and these poles therefore represent a marker for the Australian Mesoproterozoic pole path, with the Lawn Hill Platform marking the older part of the path (Idnurm 2000, p. 424). The age of the OP3 overprints are younger than *ca* 1500 Ma (Idnurm 2000).

The late Mesoproterozoic of the Albany–Fraser Belt in southwestern Australia has yielded several poles for *ca* 1.2 Ga (Pisarevsky *et al.* 2003). These are incorporated with other data in a proposed pole path for Australia spanning the interval *ca* 1.8–1.2 Ga (see below).

NEW PALEOMAGNETIC DATA

Sampling sites

A portable rock drill was used to take 2.5 cm diameter cores from most sites, although block samples were collected where water for drilling was unavailable. Most samples were oriented using a sun compass and the remainder using a magnetic compass. Strata at sites typically have low to moderate dips ($\leq 15^\circ$), and cleavage is absent.

Eighty-nine core samples were collected from the Pandurra Formation at 16 sites in Freyers railway cutting and seven block samples at a further two sites in a quarry below the water tank in the E&WS Reserve at Whyalla. Seventy core samples were collected from the Blue Range Beds at 16 sites on the coast at Talia. The locations of sections sampled are shown in Figure 2 and

Table 1 Mesoproterozoic stratigraphy and geochronology for the southern Gawler Craton–Eyre Peninsula, South Australia.

Stratigraphic unit	Lithology	Geochronology
Beda Volcanics	Amygdaloidal, massive basalt, locally brecciated and tuffaceous	Rb–Sr ages of 1.2–1.1 Ga, Sm–Nd age of *ca* 800 Ma (Cowley & Flint 1993).
Backy Point Formation	Coarse-grained sandstone, interbedded with Beda Volcanics	
Pandurra Formation	Red and light grey, coarse-grained sandstone, basal conglomerate, local shale	Whole-rock siltstone Rb–Sr isochron age of 1424 ± 51 Ma (Fanning *et al.* 1983). Youngest and dominant U–Pb detrital zircon age of 1580–1620 Ma (Fanning & Link 2003, 2004).
Blue Range Beds	Red and light grey, coarse-grained sandstone, basal conglomerate	
Charleston Granite	Massive porphyritic granite that intrudes the Corunna Conglomerate	U–Pb zircon age of 1585 ± 2 Ma (Parker & Fanning 1998)
Corunna Conglomerate	Talus breccia, medium-grained sandstone, conglomerate, local siltstone	
Roopena Volcanics	Grey-green altered amygdaloidal basalt, minor tuff	U–Pb zircon age of 1587 ± 15 Ma (Flint *et al.* 1993)
Hiltaba Granitoid Suite	Coarse-grained adamellite, granite that intrudes the Gawler Range Volcanics	U–Pb zircon age of 1589 ± 16 Ma (Creaser & Cooper 1993)
Gawler Range Volcanics	Massive, pink, porphyritic rhyolite and dacite	U–Pb zircon age of 1592 ± 3 Ma (Fanning *et al.* 1988)

Table 2 Paleomagnetic summary for the Pandurra Formation (Mean Site $\lambda_s = 33.0^\circ$S, $\phi_s = 137.6^\circ$E).

Site	N	D_h (°)	I_h (°)	k_h	α_{95} (°)	D_b (°)	I_b (°)	k_b	α_{95} (°)
PA01	4	256.3	16.3	15.6	24.0	257.0	26.2	15.6	15.6
PA02	5	250.7	33.1	10.6	24.6	251.2	43.1	10.6	24.6
PA03	3	259.5	34.3	5.84	56.5	261.4	44.0	5.84	56.5
PA04	4	253.7	20.4	32.6	16.3	254.3	30.3	32.6	16.3
PA05	3	065.0	−25.3	19.2	28.9	064.8	−35.3	19.2	28.9
PA06	6	063.8	−47.9	5.09	32.9	063.0	−57.9	5.09	32.9
PA07	4	254.2	33.9	18.0	22.3	255.3	43.8	18.0	22.3
PA08	3	239.8	48.9	25.4	25.0	237.0	58.7	25.8	24.8
PA09	5	258.5	20.7	10.7	24.5	259.1	30.6	10.7	24.5
PA10	4	264.6	32.5	25.6	18.5	264.9	42.5	25.6	18.5
PA11	4	242.6	26.3	16.1	23.6	240.5	35.7	16.1	23.6
PA12	8	253.7	37.6	15.1	14.7	247.8	46.5	15.1	14.7
PA13	7	076.1	−46.3	14.5	16.4	069.2	−54.7	14.7	16.3
PA14	5	067.4	−52.7	125.6	6.9	056.8	−59.9	125.6	6.9
PA15	3	060.2	−37.9	22.7	26.5	054.5	−45.6	22.7	26.5
PA16	11	240.5	49.0	15.5	12.0	231.9	56.6	15.5	12.0
PA17	7	247.1	32.4	9.89	20.2	251.2	45.5	9.89	20.2
PA18	5	060.7	−44.7	10.1	25.3	065.3	−58.2	10.1	25.3
Sites	18	250.3	35.3	44.0	5.3	249.2	45.2	44.2	5.3
Specimens	90	250.0	36.9	12.5	4.4	248.2	46.7	12.6	4.4

Tilt-corrected pole position: $\lambda_p = 33.6^\circ$S, $\phi_p = 64.5^\circ$E, $A_{95} = 4.9^\circ$.

N, number of specimens/sites showing ChRM; α_{95}, half-angle of the 95% confidence cone for the mean direction; D_h and I_h, *in situ* declination and inclination; D_b and I_b, declination and inclination with respect to bedding; k_h and k_b, precision parameter of Fisher (1953) distribution for mean directions *in situ* and with respect to bedding, respectively; A_{95}, 95% confidence cone for the mean virtual magnetic pole (VGP). Site mean directions are given in the sense of the predominant polarity of each site. Site details are given in Appendix 1.

details of sampling sites and rocks sampled given in Appendix 1.

Laboratory techniques

Between three and nine specimens 2.2 cm high × 2.5 cm in diameter were machined from the cores and block samples, and at least two specimens per block were processed. Routine paleomagnetic laboratory methods (Collinson 1983; Butler 1992) were employed. Remanent magnetisation was measured using a 2G 755R three-axis cryogenic magnetometer at CSIRO, North Ryde, NSW. The demagnetisers used were an in-line 2G 600 series AF demagnetisers and a Magnetic Measurements MMTD80 shielded furnace. All specimens were subjected to stepwise thermal demagnetisation with 10–15 steps.

Magnetisation components were isolated using an interactive version of Linefind (Kent *et al.* 1983), in which linear segments are fitted to data points weighted according to the inverse of their measured variances. The highest temperature to which any particular component remains stable, i.e. the highest unblocking temperature observed, 680°C, was generally taken as an indication of the Curie temperature of the mineral carrying that component.

Results for the Pandurra Formation

Our results for the Pandurra Formation are summarised in Table 2, and Figure 4 shows orthogonal projections of natural remanent magnetisation (NRM) and after thermal demagnetisation up to 680°C. Generally the structure is simple, with what appears to be a viscous component eliminated after 250–300°C, leaving a well-defined characteristic component that decays to the origin.

Directions before and after tilt-correction are shown in Figure 5. The lack of variation in structural attitudes for the Pandurra Formation where sampled did not permit a definitive fold test. The observation of mixed polarities within some specimens (see Figure 4f) agrees with chemical remanent magnetisation (CRM) acquired over a protracted interval of time, and in such circum-

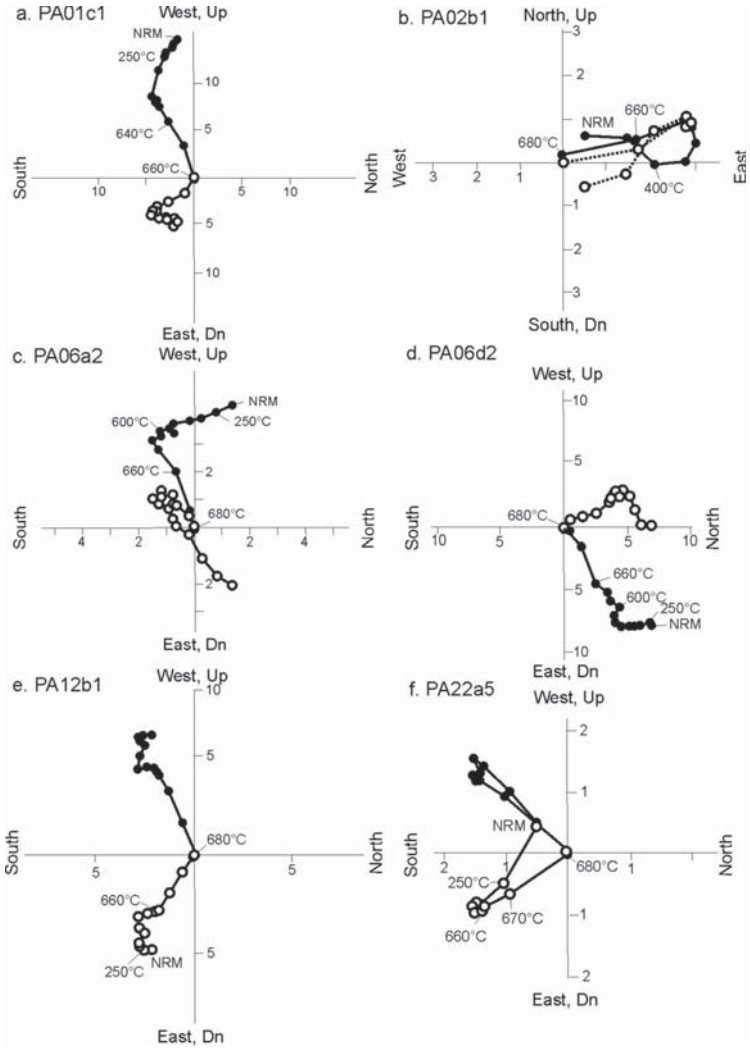


Figure 4 Orthogonal projections of remanence vectors for specimens from the Pandurra Formation after stepwise thermal demagnetisation. Solid circles plot on the horizontal plane and open circles on the vertical plane. Axes are mAm^{-1} (10^{-6} emu/cc), temperature in°C. See text for details.

stances it is appropriate to employ specimen directions, rather than site directions, to provide an overall formation mean direction (Schmidt & Williams 1995). The tilt-corrected mean direction for $N = 90$ specimens is $D = 248.2°$, $I = 46.7°$ ($\alpha_{95} = 4.4°$), indicating a paleolatitude $\lambda = 27.9 + 3.9/-3.4°$ and a pole position at latitude $\lambda_p = 33.6°$S, longitude $\phi_p = 64.5°$E ($A_{95} = 4.9°$).

Magnetostratigraphy of the Pandurra Formation

Figure 6 shows a log of the declinations of characteristic remanent magnetisation (ChRM) directions for Pandurra Formation specimens against their relative stratigraphic position. The log is in two parts because it is uncertain how the two sections are related stratigraphically (Appendix 1). The many polarity transitions at Freyers Cutting suggest several possible correlations with the two transitions observed at the quarry near the

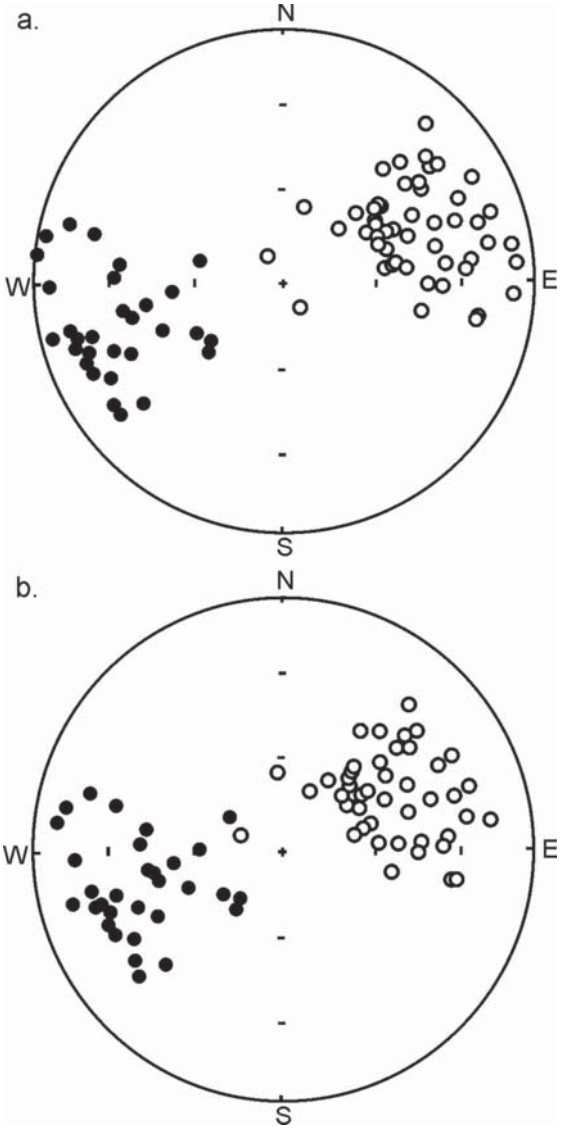


Figure 5 Stereographic projections of characteristic remanent magnetisation (ChRM) directions for the Pandurra Formation isolated by stepwise thermal demagnetisation. Solid circles plot on the lower hemisphere and open circles on the upper hemisphere: (a) present-day coordinates, and (b) after tilt correction.

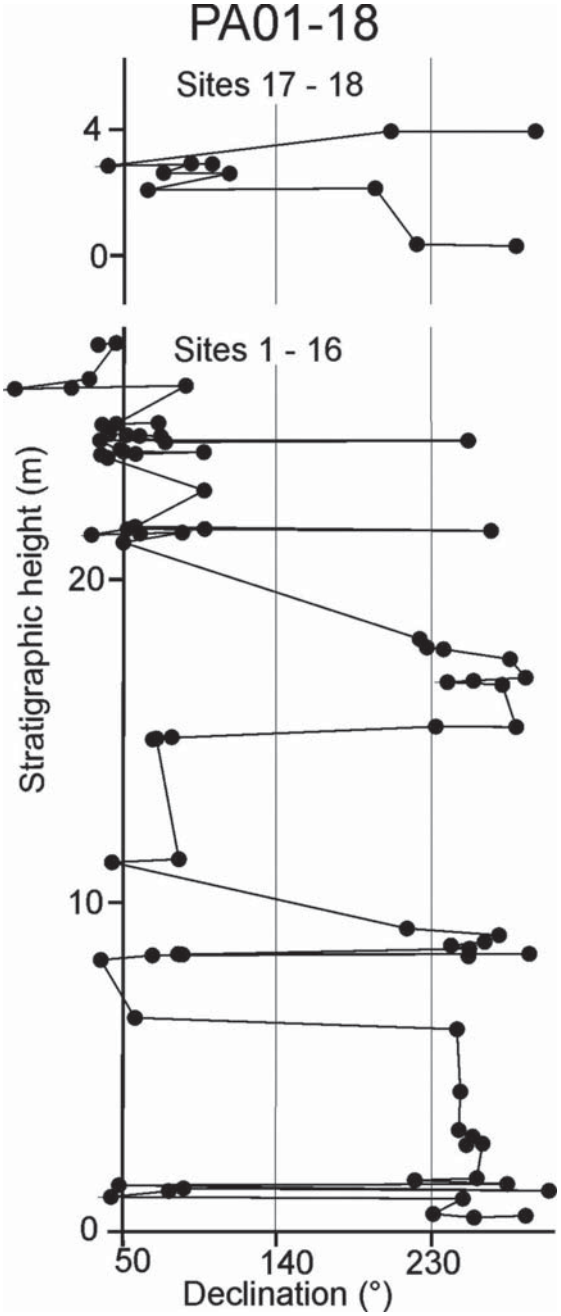


Figure 6 Magnetostratigraphy of the Pandurra Formation. Site details are given in Appendix 1. The stratigraphic relationship between sites PA01–16 and PA17–18 is uncertain, although a rough calculation places PA01–16 some 30 m below PA17–18 (see text).

water tower. Bedding at sites PA01–16 in Freyers Cutting strikes between 18° (western end) and 340° (eastern end) (true north) and dips 10°E, and sites PA17–18 from the quarry strike 316° (TN) and dip 14°NE. Projecting lines from the two localities in the directions of dip to their point of intersection, and assuming constant dips, allows the relative depths of the beds in the subsurface to be estimated. These calculations predict that the beds exposed at Freyers Cutting would occur at 650 m depth 3.7 km to the east while the quarry beds would be at 575 m depth 2.3 km NE of the quarry, where the projected lines intersect. This suggests that the quarry beds are some 30 m above the beds at Freyers Cutting. However, this extrapolation is uncertain, given the large extrapolation distances and range of strikes at Freyers Cutting. The interval sampled may span progressive early diagenesis and appears to have been magnetised over a considerable period of time incorporating about six polarity transitions, suggesting acquisition of magnetism on a timescale of 10^5–10^6 yr or more.

Results for the Blue Range Beds

Our results are given in Table 3, and Figure 7 shows orthogonal projections of end-points of NRM and after thermal demagnetisation up to 680°C. As found for the Pandurra Formation, the magnetic structure is simple, with what appears to be a viscous component eliminated after 250–300°C, leaving a well-defined characteristic component that decays to the origin.

Directions before and after tilt-correction are shown in Figure 8. As for the Pandurra Formation, the lack of variation in structural attitudes for the Blue Range Beds where sampled did not permit a definitive fold test. Additionally, mixed polarities, although comprising only two reverse directions, indicate magnetisation over an extensive period of time compared with that typifying secular variation, and it is appropriate to employ specimen directions, rather than site directions, to provide an overall formation mean direction. The tilt-corrected mean direction for $N=61$ specimens is $D=233.9°$, $I=50.2°$ ($\alpha_{95}=4.0°$), indicating a paleolatitude $\lambda=31.0+3.7/-3.5°$ and a pole position at latitude $\lambda_p=45.2°$S, longitude $\phi_p=59.0°$E ($A_{95}=4.7°$).

Combined results for the Pandurra Formation and Blue Range Beds

Although strata show low to moderate dips at the sampling sites, the Pandurra Formation dips eastwards whereas the Blue Range Beds dip to the south. These different structural attitudes therefore offer the opportunity to execute a fold test.

The fold test shows the correlation statistic (McFadden 1990) decreasing from $\xi_{2g}=52.9$ to $\xi_{2s}=2.54$ on tilt-correction (cf. with 0.05 and 0.01 significance points of 14.3 and 20.2, respectively, for $N=151$), where subscript g refers to geographic coordinates and subscript s refers to stratigraphic coordinates after tilt-correction. The fold test therefore is positive at 99% confidence and indicates that the magnetisation was acquired prior to tilting. The magnetisation may have been acquired during early diagenesis when compaction, dewatering and the partial removal of hematitic pigment likely occurred. The direction for combined data after tilt-correction ($N=151$) is $D=242.6°$, $I=48.3°$ ($\alpha_{95}=3.1°$), indicating a paleolatitude $\lambda=29.3\pm2.8°$ and a pole position at latitude $\lambda_p=38.4°$S, longitude $\phi_p=62.4°$E ($A_{95}=3.5°$).

The change in mean directions for both the Pandurra Formation and the Blue Range Beds for –40% to 140% tilt-correction is shown in Figure 9. Their closest approach before beginning to diverge is at 100% tilt-correction, in accord with the McFadden (1990) fold test described above. The inset in Figure 9 shows that the

Table 3 Paleomagnetic summary for the Blue Range Beds (Mean Site λ_s= 33.3°S, ϕ_s=134.5°E).

Site	N	D_h (°)	I_h (°)	k_h	α_{95} (°)	D_b (°)	I_b (°)	k_b	α_{95} (°)
BR01	4	64.3	−70.5	360	15.5	42.1	−59.3	36.0	15.5
BR02	2	54.3	−56.3	57.2	33.7	42.9	−44.5	57.2	33.7
BR03	4	56.7	−60.2	18.4	22.0	43.1	−48.6	18.4	22.0
BR04	4	48.3	−44.7	19.8	21.2	41.4	−32.4	19.8	21.2
BR05	4	79.2	−62.1	38.6	15.0	58.3	−54.2	38.6	15.0
BR06	4	82.9	−53.5	45.4	13.8	66.5	−47.0	45.4	13.8
BR07	4	68.5	−44.4	156	7.4	58.4	−35.3	156	7.4
BR08	4	86.3	−63.6	70.2	11.0	62.5	−56.8	70.2	11.0
BR09	4	86.8	−65.1	8.37	33.8	61.4	−58.3	8.37	33.8
BR10	4	102.6	−61.4	46.1	13.7	76.7	−58.6	46.1	13.7
BR11	4	74.1	−55.0	155	7.4	58.7	−46.6	155	7.4
BR12	4	95.3	−54.4	29.7	17.2	76.1	−50.6	29.7	17.2
BR13	4	59.4	−65.6	18.2	22.2	42.3	−54.1	18.2	22.2
BR14	4	45.6	−59.6	72.9	10.8	35.4	−46.6	72.9	10.8
BR15	3	55.1	−58.5	82.6	13.7	42.7	−46.7	82.6	13.7
BR16	4	71.7	−59.1	10.4	30.0	54.7	−50.0	10.4	30.0
Sites	16	70.1	−59.4	51.7	5.2	53.3	−50.0	51.7	5.2
Specimens	61	71.0	−59.5	21.6	4.0	53.9	−50.2	21.6	4.0

Tilt-corrected pole position: λ_p = 45.2°S, ϕ_p = 59.0°E, A_{95} = 4.7°. See Table 2 for explanation of column headings and conventions. Site details are given in Appendix 1.

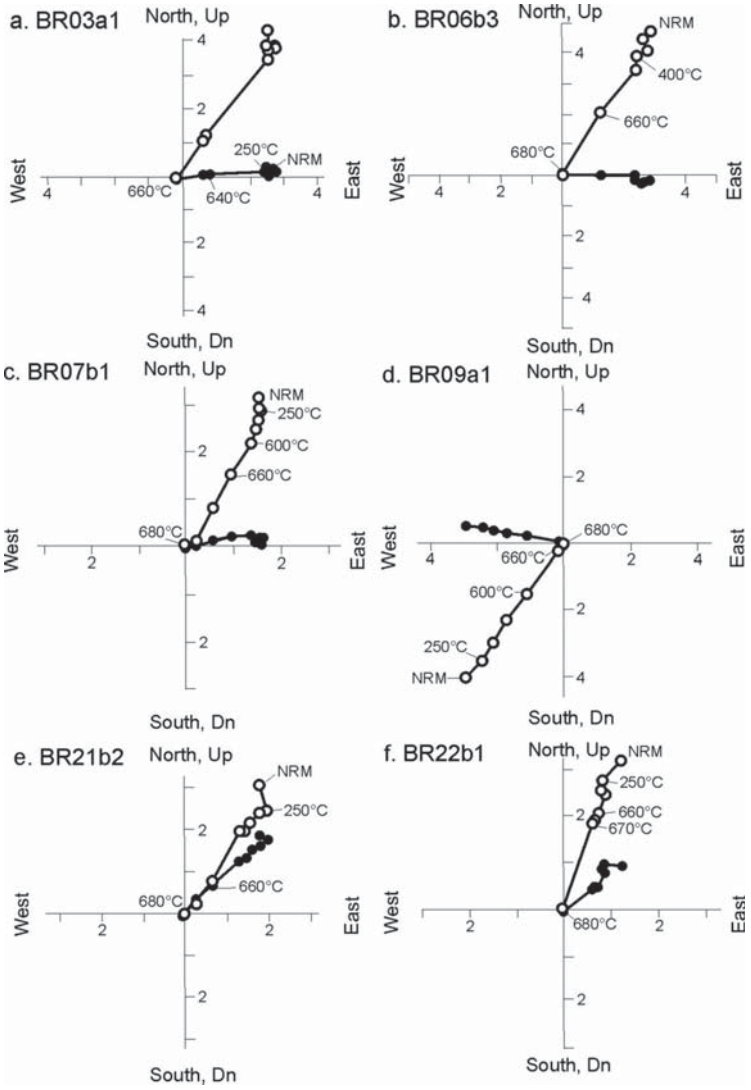


Figure 7 Orthogonal projections of remanence vectors for specimens from the Blue Range Beds after stepwise thermal demagnetisation. Solid circles plot on the horizontal plane and open circles on the vertical plane. Axes are mAm^{-1} (10^{-6} emu/cc), temperature in°C. See text for details.

maximum for the precision parameter, k, occurs at 100% tilt-correction, indicating that these units likely acquired their magnetisation when their bedding was horizontal.

DISCUSSION

Mesoproterozoic pole path for Australia

The pole positions from the Pandurra Formation and Blue Range Beds are plotted on a Mesoproterozoic APWP for Australia (Figure 10), modified after Idnurm & Giddings (1988) and Idnurm (2000). The combined pole position for the two units is also plotted. Data for the APWP are summarised in Table 4.

The mean pole position for the Pandurra Formation and Blue Range Beds plots after the OP3 group of overprint poles for the McArthur Basin and Lawn Hill Platform, which are younger than *ca* 1.5 Ga, and before the Albany–Fraser orogen poles of *ca* 1.2 Ga. This is consistent with the inferred Mesoproterozoic age for the Pandurra Formation and Blue Range Beds based on stratigraphy and geochronology.

Comparison with the North American Mesoproterozoic pole path

The Australian Mesoproterozoic pole path shown in Figure 11 is compared with the North American pole path for the Mesoproterozoic in North American coordinates after rotating Australia about Euler poles depending on polarity. In Figure 11a the Euler pole used was latitude $\lambda_e = 13^\circ$ S, longitude $\phi_e = 139^\circ$ E and angle $\omega_e = 120^\circ$, and for Figure 11b the Euler pole was latitude $\lambda_e = 5^\circ$ S, longitude $\phi_e = 260^\circ$ E and angle $\omega_e = 100^\circ$.

In Figure 11a the paths converge at 1.4 Ga and remain in close proximity until diverging at 1.2 Ga. It is unclear if this indicates that Australia and North America were together in some supercontinent for that interval. Based on this comparison the continents were likely very widely separated at that time. Data of Pisarevsky *et al.*

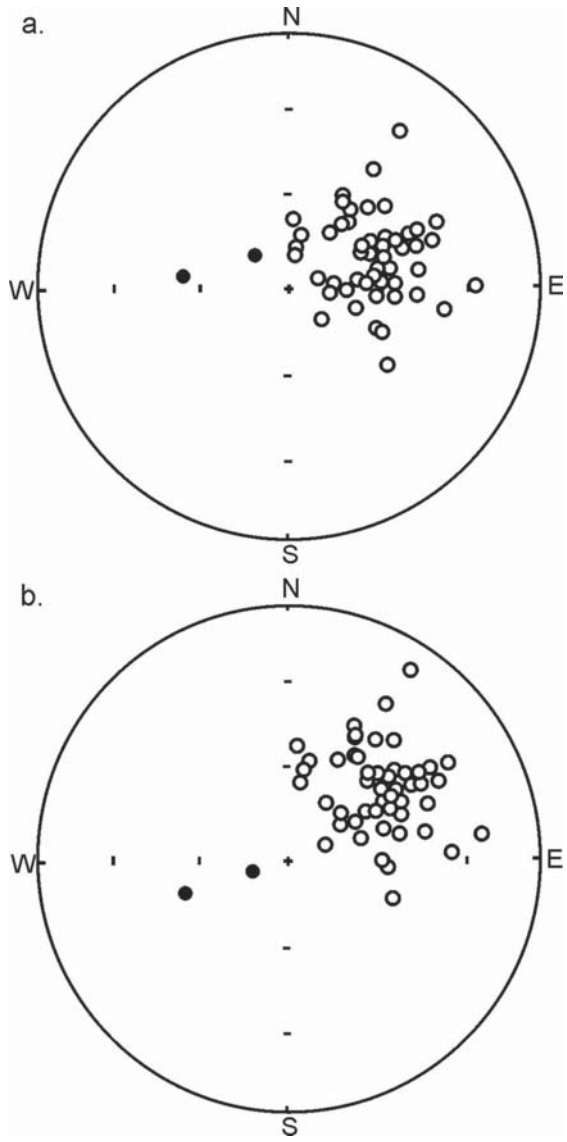


Figure 8 Stereographic projections of characteristic remanent magnetisation (ChRM) directions for the Blue Range Beds isolated by stepwise thermal demagnetisation. Solid circles plot on the lower hemisphere and open circles on the upper hemisphere: (a) present-day coordinates; (b) after tilt correction.

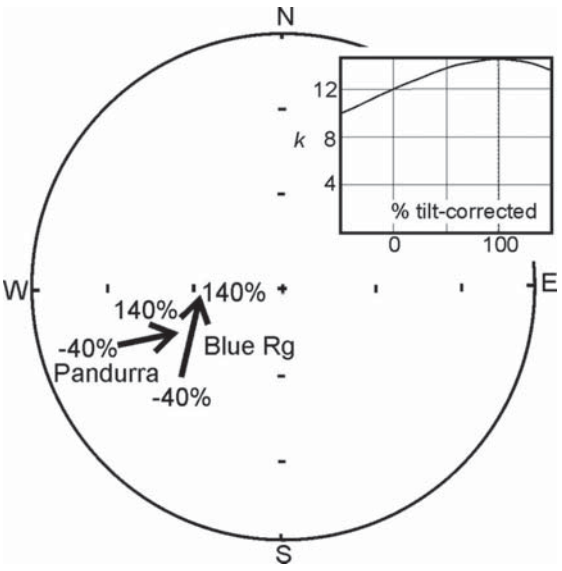


Figure 9 Stereographic projection showing the changing mean directions for the Pandurra Formation and Blue Range Beds on restoring them to the horizontal. This is performed from –40% to 140% tilt-correction to allow for any statistical aberration. The inset shows that the precision parameter kappa, k, peaks at 100% tilt-correction, suggesting that the magnetisation was acquired when both units were horizontal, consistent with the McFadden (1990) fold test result (see text).

(2003, p. F8) place 'Australia at high latitudes at *ca* 1.2 Ga, whereas existing data show that Laurentia occupied low latitudes.' This militates against any such supercontinent and suggests the close proximity of the paths at 1.4–1.2 Ga is fortuitous.

The paths in Figure 11b, for the opposite polarites to that shown in Figure 11a, show greater similarities than those in Figure 11a, but the presence of differences suggest that the comparison is of little significance. First, it is thought that the McArthur Basin–Lawn Hill OP3 overprints are *ca* 1.5 Ga, whereas the Western Channel diabase is closer to 1.6 Ga. These two poles plot quite closely, being some distance from the western Pacific location for Australia's 1.6 Ga pole position (Figure 10). Nevertheless, the paths seem to merge sometime before *ca* 1.4 Ga, when they appear to be in good agreement. The paths then track semi-parallel to each other until *ca* 1.2 Ga, as for the other polarity option, after which they diverge (cf. Figure 11a).

For this second configuration Australia and North America are even further separated, and seem to occupy different hemispheres for this period. On balance our preferred interpretation is that mid-Mesoproterozoic APWPs for Australia and North America show similarities, but it is unlikely these continents were part of a supercontinent from 1.4 to 1.2 Ga.

CONCLUSIONS

The fluvial red bed Pandurra Formation and Blue Range Beds of the Gawler Craton, South Australia, yield comparable pre-tilting magnetisations isolated after thermal demagnetisation. The steeply dipping Blue Range Beds reported from the Ningana area could not be located during this study, which prevented an internal fold test to be executed for that unit. However, taken together the directions from both units yield a positive fold test. The apparent ages of the respective pole positions and their mean pole position, when compared with Australia's Mesoproterozoic APWP, imply apparent ages between 1.5 and 1.2 Ga, consistent with the inferred Mesoproterozoic depositional ages of the two units.

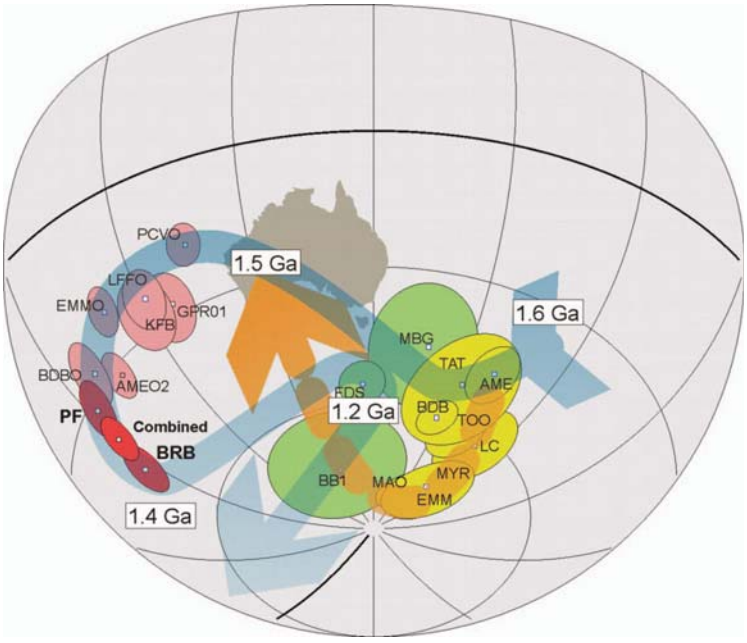


Figure 10 Mesoproterozoic apparent polar wander path for Australia. See Table 4 for details of paleomagnetic poles. The positions of the poles for the Pandurra Formation (PF), the Blue Range Beds (BRB) and the combined pole imply apparent ages between 1.5 and 1.2 Ga, in accord with the inferred stratigraphic ages of the two units. The blue pole path short-circuits the more southerly excursion as originally published by Idnurm (2000), shown in orange. We believe that poles for the latter path are strongly biased by the effects of Tertiary weathering; see Schmidt & Williams (2008, p. 277) for more detailed argument.

Table 4 Summary of selected late Paleoproterozoic and Mesoproterozoic poles for Australia (quality factor Q at least 5/7 for ChRM and 4/7 for overprints).

Formation	Age (Ga)	Dec (°)	Inc (°)	α_{95} (°)	λ_p (°)	ϕ_p (°)	dp† (°)	dm (°)	1	2	3	4	5	6	7	Q	Ref.
BB1	*ca* 1.2	178.2	67.0	8.9	−74.4	123.8	12.2	14.7	×	√	√	×	√	√	√	5	a
MBG	*ca* 1.205	124.0	68.8	8.2	−46.6	167.4	11.9	13.9	×	√	√	×	√	√	√	5	a
FDS	*ca* 1.212	152.6	74.1	2.9	−55.8	145.7	4.7	5.2	√	√	×	+C	√	√	√	6	a
Blue Range Beds	>1.1<1.6	233.9	50.2	4.0	−45.2	59.0	4.7		×	√	√	×	√	√	√	5	herein
Pandurra Formation	>1.1<1.6	248.3	46.7	4.4	−33.6	64.5	4.9		×	√	√	×	√	√	√	5	herein
BRB & PF comb.	>1.1<1.6	242.6	48.3	3.1	−38.4	62.4	3.5		×	√	√	√	√	√	√	6	herein
AMEO2	*ca* 1.5	242.8	51.7	3.6	−32.2	74.8	4.4		×	√	√	×	√	×	√	4	b
BDBO	*ca* 1.5	247.6	44.7	5.1	−27.8	68.8	5.4		×	√	√	×	√	×	√	4	b
EMMO	*ca* 1.5	256.6	56.3	3.4	−21.9	81.8	4.4		×	√	√	×	√	×	√	4	b
KFB		245.5	65.6	4.8	−27.1	93.7	6.8		×	√	√	×	√	√	√	5	c
LYNO		*255.7*	*54.4*	*5.3*	*−21.5*	*78.3*	*5.9*		×	×	√	×	√	×	√	*3*	*d*
LWNO1		*234.7*	*66.9*	*4.8*	*−37.1*	*97.3*	*7.7*		×	×	√	×	√	×	√	*3*	*d*
FSHO2		*262.1*	*70.5*	*5.6*	*−18.9*	*101.5*	*9.3*		×	×	√	×	√	×	√	*3*	*d*
LLFO (last 3)	*ca* 1.5	253.0	65.0	3.8	−24.5	93.4	3.4		×	√	√	×	√	×	√	4	herein
GPRO1	*ca* 1.5	250.0	66.8	4.4	−27.4	97.1	6.5		×	√	√	–F	√	×	√	5	d
PCVO		265.6	72.7	2.5	−17.5	105.3	4.2		×	√	√	×	√	√	√	5	d
MALO		*351.3*	*−38.2*	*8.9*	*−80.7*	*193.4*	*9.8*		×	×	√	×	√	×	√	*3*	*b*
AMEO1		*350.2*	*−40.9*	*5.7*	*−81.8*	*156.6*	*6.9*		×	×	√	*+F*	√	×	√	*4*	*b*
MAO (last 2)		350.6	−39.8	4.8	−81.7	172.5	5.0		×	√	√	+F	√	×	√	5	herein
EMM		349.1	−36.4	5.7	−79.1	202.6	6.1		×	√	√	+F	√	×	√	5	b
MYR		346.5	−38.6	8.2	−75.9	197.4	7.7		×	√	√	+F	√	×	√	5	b
TOO		334.3	−50.1	5.2	−61.0	186.7	6.1		×	√	√	+F	√	√	√	6	b
BDB		336.1	−55.6	3.3	−60.6	176.1	4.3		×	√	√	+F	√	√	√	6	b
LCG		152.0	50.0	7.2	−63.0	201.0	9.0		×	√	√	×	√	√	√	5	e
TAT		326.5	−56.4	9.9	−52.7	182.2	10.7		×	√	√	+F	√	×	√	5	b
AME		318.0	−55.1	4.9	−47.4	188.8	6.1		×	√	√	+F	√	×	√	5	b

dp, dm, semi-axes of error ellipse around the pole of probability of 95%; see Table 2 for explanation of other column headings.
† where only one entry is given under dp, dm the value is for A_{95}.
1–7 and Q, Reliability criteria of Van der Voo (1990). a, Pisarevsky *et al.* (2003); b, Idnurm *et al.* (1995); c, Giddings & Idnurm (1993). +F positive fold test, –F negative fold test, sF synfolding, +C positive conglomerate test.
Rows in italics have been combined in the immediately following row (indicated as 'herein') so the minimum values for criterion number 2 of Van der Voo (1990) ($N > 24$, $k > 10$, $\alpha < 16°$) are met.

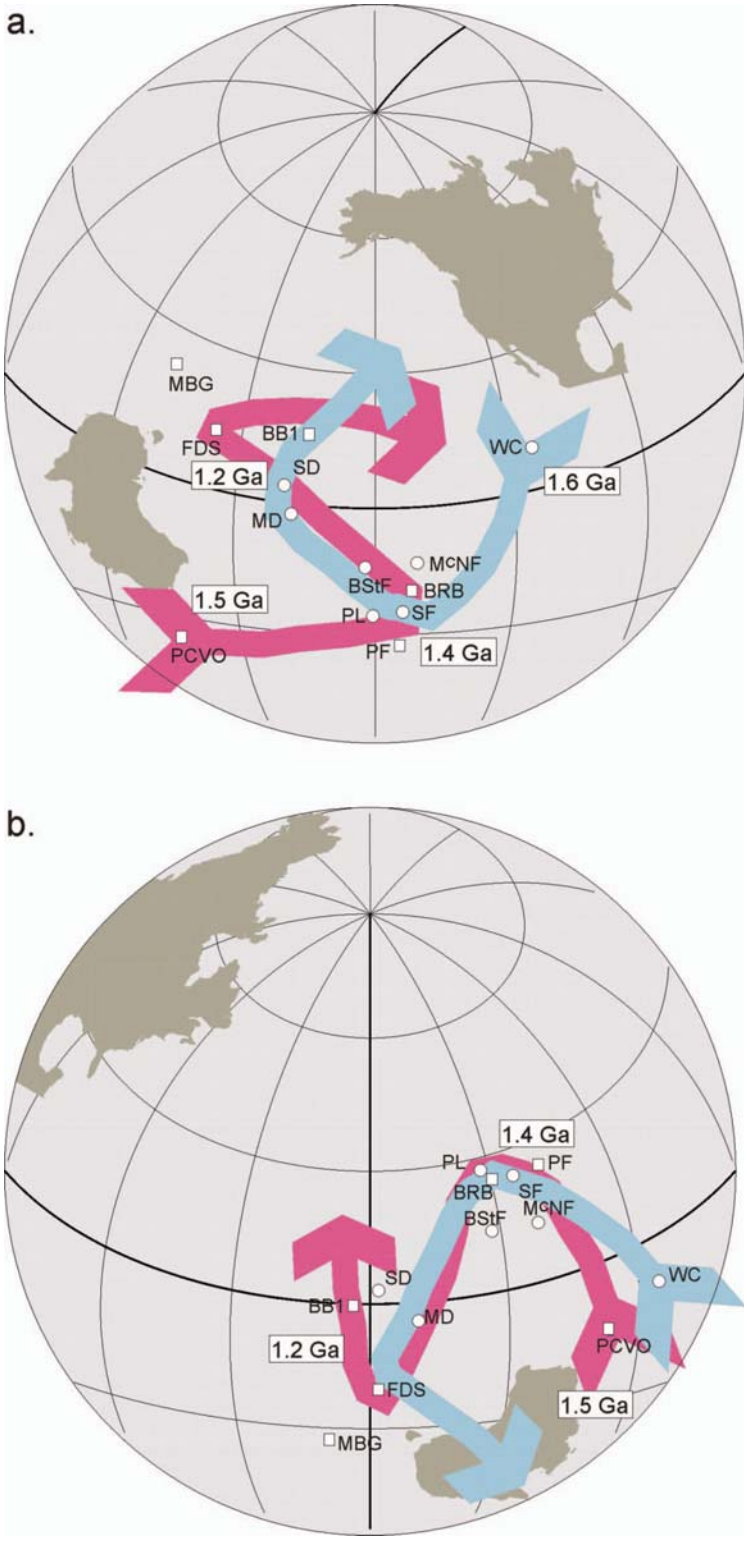


Figure 11 Comparisons using both polarities of Australia's Mesoproterozoic apparent polar wander path with a similarly aged segment of the North American (NA) apparent polar wander path (in NA coordinates). Euler poles were arrived at empirically, to overlay the relevant pole paths. In Figure 11a, the 'normal' polarity NA path (blue) is compared with the Australian (red) path, without specifying whether it is 'normal' or 'reverse'—the Euler pole used to rotate Australia and its poles to NA was latitude $\lambda_e = 13^\circ$ S, longitude $\phi_e = 139^\circ$ E and angle $\omega_e = 120^\circ$ (anticlockwise). In Figure 11b, we keep the same polarity Australian (red) path and compare with the 'reverse' polarity NA path (blue)—the Euler pole used to rotate Australia and its poles to NA was latitude $\lambda_e = 5^\circ$ S, longitude $\phi_e = 260^\circ$ E and angle $\omega_e = 100^\circ$ (anticlockwise). See Figure 10 and Table 4 for the Australian poles. The North American poles are: SD, Sudbury dykes (GPMDB Refno. 1097 and 1879); MD, Mackenzie Dykes (Buchan & Halls 1990); McNF, McNamara Formation (GPMDB Refno. 3556); PL, Purcell lava (GPMDB Refno. 3556), Snowslip Formation (GPMDB Refno. 3556); WC, Western Channel diabase (GPMDB Refno. 293) (Hamilton & Buchan 2007).

Comparing similar aged pole positions for Mesoproterozoic North America shows some similarities after applying appropriate Euler rotations to the Australian poles, depending on the magnetic polarity adopted. For both polarity options the paths are sub-parallel from about 1.4 to 1.2 Ga, before diverging. During this interval Australia and North America were widely separated, and in one configuration even occupied

opposite hemispheres, which would rule out their assemblage in a supercontinent reconstruction for this period. The Mesoproterozoic may have been a time during which continents drifted as separate entities or in smaller clusters of continents.

ACKNOWLEDGEMENTS

We thank Clive Foss and Mark Lackie for reviews of the draft manuscript which significantly improved the final version.

REFERENCES

Buchan K. L. & Halls H. C. 1990. Paleomagnetism of Proterozoic mafic dyke swarms of the Canadian Shield. *In*: Parker A. J., Rickwood P. C. & Tucker D. H. eds. *Mafic dykes and emplacement mechanisms.* Balkema, 209–230.

Butler R. F. 1992. *Paleomagnetism: magnetic domains to geologic terranes.* Blackwell, Oxford, 319 pp.

Chamalaun F. H. & Dempsey C. E. 1978. Palaeomagnetism of the Gawler Range Volcanics and implications for the genesis of the Middleback hematite orebodies. *Journal Geological Society of Australia* **25**, 255–265.

Collinson D. W. 1983. *Methods in rock magnetism and palaeomagnetism.* Chapman & Hall, London, 503 pp.

Cowley W. M. 1993. Cariewerloo Basin. *In*: Drexel J. F., Preiss W. V. & Parker A. J. eds. *The geology of South Australia, Volume 1, the Precambrian. Geological Survey of South Australia Bulletin* **54**, 139–142.

Cowley W. M. & Flint R. B. 1993. Epicratonic igneous rocks and sediments. *In*: Drexel J. F., Preiss W. V. & Parker A. J. eds. *The Geology of South Australia, Volume 1, The Precambrian. Geological Survey of South Australia Bulletin* **54**, 142–147.

Creaser R. A. & Cooper J. A. 1993. U–Pb geochronology of middle Proterozoic felsic magmatism surrounding the Olympic Dam Cu–U–Au–Ag and Moonta Cu–Au–Ag deposits, South Australia. *Economic Geology* **88**, 186–197.

Fanning C. M. & Link P. K. 2003. Detrital zircon provenance of the Mesoproterozoic Pandurra Formation, South Australia: Gawler Craton zircon population and implications for the Belt Supergroup. *Geological Society of America Abstracts with Programs* **35** (6), 465.

Fanning C. M. & Link P. K. 2004. Detrital zircon provenance of the Mesoproterozoic Pandurra Formation, South Australia: Gawler Craton derived age spectra and implications for the source of the Belt Supergroup, USA. *Geological Society of Australia Abstracts* **17**, 155.

Fanning C. M., Flint R. B. & Preiss W. V. 1983. Geochronology of the Pandurra Formation. *Geological Survey of South Australia Quarterly Geological Notes* **88**, 11–16.

Fanning C. M., Flint R. B., Parker A. J., Ludwig K. R. & Blissett A. H. 1988. Refined Proterozoic evolution of the Gawler Craton, South Australia, through U–Pb zircon geochronology. *Precambrian Research* **40/41**, 363–386.

Fisher R. 1953. Dispersion on a sphere. *Proceedings of the Royal Society of London* **A217**, 295–305.

Flint R. B. 1992. *Elliston, South Australia.* Department of Mines and Energy South Australia, 1:250,000 Geological Series, Explanatory Notes Sheet SI 53-6, 40 pp.

Flint R. B. 1993. Itiledoo Basin. *In*: Drexel J. F., Preiss W. V. & Parker A. J. eds. *The geology of South Australia, Volume 1, the Precambrian. Geological Survey of South Australia Bulletin* **54**, 142.

Flint R. B. & Parker A. J. 1981. The Blue Range Beds, central Eyre Peninsula. *Geological Survey of South Australia Quarterly Geological Notes* **80**, 12–15.

Flint R. B. & Rankin L. R. 1989. *Kimba map sheet SI 53-7, 1:250 000 Geological Series.* Department of Mines and Energy South Australia, Adelaide.

Flint R. B. & Rankin L. R. 1991. *Kimba, South Australia.* Department of Mines and Energy South Australia, Adelaide, 1:250 000 Geological Series, Explanatory Notes Sheet SI 53-7, 46 pp.

Flint R. B., Blissett A. H., Conor C. H. H., Cowley W. M., Cross K. C., Creaser R. A., Daly S. J., Krieg G. W., Major R. B., Teale G. S. & Parker A. J. 1993. Mesoproterozoic. *In*: Drexel J. F., Preiss W. V. & Parker A. J. eds. *The geology of South Australia, Volume 1, the Precambrian. Geological Survey of South Australia Bulletin* **54**, 106–169.

Giddings J. W. & Idnurm M. 1993. Significance of overprint magnetizations in the Palaeoproterozoic Kombolgie Formation, western McArthur Basin, N.T. *Exploration Geophysics* **24**, 231–238.

Hamilton M. A. & Buchan K. L. 2007. U–Pb geochronology of the Western Channel diabase, Wopmay orogen: Implications for the APWP for Laurentia in the earliest Mesoproterozoic. *Geological Association of Canada Abstracts* **32**, 35–36.

Idnurm M. 2000. Towards a high resolution late Palaeoproterozoic–earliest Mesoproterozoic apparent polar wander path for northern Australia. *Australian Journal of Earth Sciences* **47**, 405–429.

Idnurm M. & Giddings J. W. 1988. Australian Precambrian polar wander: a review. *Precambrian Research* **40/41**, 61–88.

Idnurm M., Giddings J. W. & Plumb K. A. 1995. Apparent polar wander and reversal stratigraphy of the Palaeo–Mesoproterozoic southeastern McArthur Basin, Australia. *Precambrian Research* **72**, 1–41.

Irving E. & Green R. 1958. Polar movement relative to Australia. *Geophysical Journal of the Royal Astronomical Society* **1**, 64–72.

Kent J. T., Briden J. C. & Mardia K. V. 1983. Linear and planar structure in ordered multivariate data as applied to progressive demagnetization of palaeomagnetic remanence. *Geophysical Journal of the Royal Astronomical Society* **75**, 593–621.

Kohn B. P., Gleadow A. J. W., Brown R. W., Gallagher K., O'Sullivan P. B. & Foster D. A. 2002. Shaping the Australian crust over the last 300 million years: insights from fission track thermotectonic imaging and denudation studies of key terranes. *Australian Journal of Earth Sciences* **49**, 697–717.

McFadden P. L. 1990. A new fold test for palaeomagnetic studies. *Geophysical Journal International* **103**, 163–169.

Parker A. J. 1993. Kimban Orogeny. *In*: Drexel J. F., Preiss W. V. & Parker A. J. eds. *The Geology of South Australia, Volume 1, The Precambrian. Geological Survey of South Australia Bulletin* **54**, 71–81.

Parker A. J. & Fanning C. M. 1998. *Whyalla, South Australia.* Primary Industries and Resources SA, 1:250 000 Geological Series, Explanatory Notes Sheet SI 53-8, 52 pp.

Parker A. J. & Lemon N. M. 1982. Reconstruction of the Early Proterozoic stratigraphy of the Gawler Craton, South Australia. *Journal of the Geological Society of Australia* **29**, 221–238.

Pisarevsky S. A., Wingate M. T. D. & Harris L. B. 2003. Late Mesoproterozoic (*ca* 1.2 Ga) palaeomagnetism of the Albany–Fraser orogen: no pre-Rodinia Australia–Laurentia connection. *Geophysical Journal International* **155**, F6–F11.

Preiss W. V. (Compiler) 1987a. The Adelaide Geosyncline. Late Proterozoic stratigraphy, sedimentation, palaeontology and tectonics. *Geological Survey of South Australia Bulletin* **53**, 438 pp.

Preiss W. V. 1987b. Basement to the Adelaide Geosyncline. *In*: Preiss W. V. compiler. The Adelaide geosyncline. Late Proterozoic stratigraphy, sedimentation, palaeontology and tectonics. *Geological Survey of South Australia Bulletin* **53**, 35–41.

Preiss W. V. 1993. Neoproterozoic. *In*: Drexel J. F., Preiss W. V. & Parker A. J. eds. *The geology of South Australia, Volume 1, the Precambrian. Geological Survey of South Australia Bulletin* **54**, 170–203.

Schmidt P. W. & Clark D. A. 2011. Magnetic characteristics of the Hiltaba Suite Granitoids and Volcanics: late Devonian overprinting and thermal history of the Gawler Craton. *Australian Journal of Earth Sciences* **58**, 361–374.

Schmidt P. W. & Embleton B. J. J. 1985. Prefolding and postfolding magnetic signatures in Precambrian (~ 2.9–2.7 Ga) igneous rocks from the Pilbara Craton and Hamersley Basin, NW Australia. *Journal of Geophysical Research* **90**, 2967–2984.

Schmidt P. W. & Williams G. E. 1995. The Neoproterozoic climatic paradox: Equatorial palaeolatitude for Marinoan glaciation near sea level in South Australia. *Earth and Planetary Science Letters* **134**, 107–124.

SCHMIDT P. W. & WILLIAMS G. E. 2008. Palaeomagnetism of red beds from the Kimberley Group, Western Australia: implications for the palaeogeography of the 1.8 Ga King Leopold glaciation. *Precambrian Research* **167**, 267–280.

STRIK G., BLAKE T. S., ZEGERS T. E., WHITE S. H. & LANGEREIS C. G. 2003. Palaeomagnetism of flood basalts in the Pilbara Craton, Western Australia: late Archaean continental drift and the oldest known reversal of the geomagnetic field. *Journal of Geophysical Research*, **108** (B12), 2551, doi:10.1029/2003JB002475.

VAN DER VOO R. 1990. The reliability of paleomagnetic data. *Tectonophysics* **184**, 1–9.

WEBB A. W., THOMSON B. P., BLISSETT A. H., DALY S. J., FLINT R. B. & PARKER A. J. 1986. Geochronology of the Gawler Craton, South Australia. *Australian Journal of Earth Sciences* **33**, 119–143.

WILLIAMS G. E. 1994. Acraman: A major impact structure from the Neoproterozoic of Australia. *In*: Dressler, B. O., Grieve, R. A. F. & Sharpton, V. L. eds. *Large meteorite impacts and planetary evolution*. Geological Society of America Special Paper 293, 209–224.

WILLIAMS G. E. 2001. Neoproterozoic (Torridonian) alluvial fan succession, northwest Scotland, and its tectonic setting and provenance. *Geological Magazine* **138**, 471–494.

WILSON T., BOSMAN S., HEATH P., GOUTHAS G., COWLEY W., MAUGER A., BAKER A., GORDON G., DHU T., FAIRCLOUGH M. & DELANEY G. 2010. The search for unconformity-related uranium mineralisation in the Pandurra Formation, South Australia: an international multidisciplinary collaboration. *MESA Journal* **58**, 9–15.

WINGATE M. T. D., CAMPBELL I. H., COMPSTON W. & GIBSON G. M. 1998. Ion-probe U–Pb ages for Neoproterozoic basaltic magmatism in south-central Australia and implications for the breakup of Rodinia. *Precambrian Research* **87**, 135–159.

ZANG W-L. & FANNING C. M. 2001. Age of the Kimban Orogeny revealed—U–Pb dates on the Corny Point Paragneiss, Yorke Peninsula. *MESA Journal* **23**, 28–33.

ZHAO J-X., MCCULLOCH M. T. & KORSCH R. J. 1994. Characterization of a plume-related ~800 Ma magmatic event and its implications for basin formation in central-southern Australia. *Earth and Planetary Science Letters* **121**, 349–367.

Received 14 October 2010; accepted 23 February 2011

APPENDIX 1: SAMPLING SITES

Pandurra Formation

Sites PA01–16. Freyers railway cutting, Whyalla, from 33°00′59″S, 137°33′12″E (site PA01) to 33°01′02″S, 137°33′17″E (site PA21 *ca* 160 m east of site PA01) (Whyalla 1:100 000 topographic map). Pale to moderate red and very light grey, cross-bedded coarse-grained sandstone. Strike 18° (western end) to 340° (eastern end) (true north), dip 10°E, 27.5 m section thickness.

Sites PA17–18. Quarry below the water tank in the E&WS Reserve, Whyalla, 33°01′17″S, 137°33′16″E (Whyalla 1:100 000 topographic map). Pale to moderate red, coarse-grained sandstone. Strike 316° (TN), dip 14°NE, 4 m section thickness.

Blue Range Beds

Sites BR01–04. Section along Mount Camel Beach, 33°17′00″S, 134°44′56″E (Elliston 1:250 000 topographic map). Pale to moderate red and very light grey, cross-bedded coarse-grained sandstone. Strike 97° (TN), dip 15°S, 102 m section thickness.

Sites BR05–07. Coastal section at Woolshed Cave, 33°18′25″S, 134°45′50″E (Elliston 1:250 000 topographic map). Pale to moderate red and very light grey, cross-bedded coarse-grained sandstone. Strike 107° (TN), dip 15°SW, 6.1 m section thickness.

Sites BR08–10. Coastal section at The Tub, 33°18′50″S, 134°46′31″E (Elliston 1:250 000 topographic map). Pale to moderate red and very light grey, cross-bedded coarse-grained sandstone. Strike 107° (TN), dip 15°SW, 4.2 m section thickness.

Sites BR11–16. Coastal section at The Monument, 33°19′39″S, 134°47′55″E (Elliston 1:250 000 topographic map). Pale to moderate red and very light grey, cross-bedded coarse-grained sandstone, with some heavy-mineral bands. Strike 107° (TN), dip 15°SW, 19.4 m section thickness.

Copyright of Australian Journal of Earth Sciences is the property of Taylor & Francis Ltd and its content may not be copied or emailed to multiple sites or posted to a listserv without the copyright holder's express written permission. However, users may print, download, or email articles for individual use.